LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	29
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Directional static stability, prop effects, damping in Yaw, directional control, Effect of wing sweep back, spin, lateral dynamic stability.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Directional static stability, prop effects, damping in Yaw, directional control, Effect of wing sweep back, spin, lateral dynamic stability.
5. Specific Objectives	6.1.4.1 Trainee will be able to describe aircraft stability in all the three axis.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of flight characteristics diagrams; Instructor supervision required for theoretical application.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define the three axes of aircraft motion.
(b) What is the primary function of the rudder?
(c) Recall the definition of static stability.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Directional static stability definition • Propeller effects on yaw • Damping in Yaw mechanisms • Directional control surfaces • Effect of wing sweep back	L & Ds	PPT, IB	15 min
(b)	• Spin characteristics and recovery • Lateral dynamic stability • Dutch roll phenomenon • Phugoid motion overview • Fixed wing autorotation	L & Ds	PPT, PDF	20 min
(c)	• Summary of stability axes • Integration of yaw and roll • Preparation for dynamic stability • Review of specific objectives • Link to next lesson plan	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how wing sweep affects directional stability.
(b) Differentiate between static and dynamic stability in yaw.
(c) Describe the role of damping in preventing oscillation.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from static to dynamic stability concepts to be discussed in LP 30.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Dynamic stability: longitudinal and lateral, Phugoid, Dutch roll, Fixed wing autorotation. Short period oscillation, spiral divergence and Directional divergence.	Refer to the master syllabus reference list	LP 30

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min